Intelligent Fall Monitoring System for Elderly Care

INDIVIDUAL THESIS

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ABSTRACT

Falls among older adults are a major healthcare concern because they can lead to serious injuries, reduced quality of life, and increased healthcare costs. Many existing fall detection systems rely primarily on motion sensors and can only detect a fall after it has occurred. However, physiological changes such as abnormal heart rate and blood pressure fluctuations often appear before a fall, particularly in cases related to dizziness, fainting, or hypotension. This research presents a low-cost wearable Photoplethysmography (PPG)-based biometric fall risk prediction system designed as a component of an integrated elder-care fall detection solution. The system utilizes a MAX30102 PPG sensor and an ESP32 microcontroller to continuously collect physiological data and calculate heart rate parameters. A personalized dataset was collected using the developed device together with a standard blood pressure cuff as the reference measurement tool. A Random Forest machine learning model was trained to estimate systolic and diastolic blood pressure from PPG-derived features and heart rate measurements. The predicted blood pressure and heart rate values were then used to classify fall risk through clinically informed threshold-based rules. The developed solution transmits data in real time through MQTT using a HiveMQ cloud broker, while a Python backend performs blood pressure prediction and provides live visualization through a React dashboard. Experimental evaluation showed that the personalized model achieved a systolic blood pressure Mean Absolute Error (MAE) of 4.91 mmHg and a diastolic MAE of 3.46 mmHg under similar testing conditions. The study demonstrates the feasibility of integrating continuous physiological monitoring into a fall detection framework to provide early warning capabilities and support improved fall risk assessment. Although the system represents a proof of concept and requires validation with larger populations, it highlights the potential of personalized PPG-based monitoring for proactive elderly care.

Keywords: Photoplethysmography (PPG), Blood Pressure Prediction, Fall Risk Assessment, Machine Learning, Elderly Care

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
PPG	photoplethysmography
BP	Blood Pressure
ICU	intensive care units
MAE	Mean Absolute Error
ML	Machine Learning
AI	Artificial Intelligence
HR	Heart Rate
BPM	Beats Per Minute
SpO_2	Peripheral Oxygen Saturation

1 INTRODUCTION

1.1 Overall project context

Falls are one of the main health risks for elderly people. A fall does not usually happen without warning. Before some falls, such as those caused by fainting or dizziness, the body shows signs like a drop in blood pressure or a change in heart rate. This project builds an elder-care fall detection system with four components: (1) a motion-based fall detector using accelerometer (x, y, z) data, (2) an environmental sensing component, (4) an alert component, and (3) my component, a PPG-based biometric fall risk prediction system. Together, these components detect falls more reliably than a single sensor.

By continuously monitoring physiological instabilities prior to an event, my component cross-validates the motion and environmental detection outputs, thereby significantly reducing false alarms and enhancing the overall system accuracy.

1.2 Individual component context

My component is the **PPG-Based Biometric Fall Risk Prediction System**. It uses a MAX30102 sensor, which shines red and infrared light into the fingertip and measures how much light the blood absorbs. This gives raw photoplethysmography (PPG) signals. An ESP32 microcontroller reads these signals, finds the heartbeat peaks, and calculates the heart rate (BPM) and the pulse interval (the time between two heartbeats). The ESP32 sends this data in JSON format to a HiveMQ MQTT broker in the cloud.

The Python backend subscribes to the broker and receives the data. It predicts blood pressure (BP) using a machine learning model, saves the data in the database, and shows it on the React dashboard in real time. The model is trained on a personalised dataset, which was collected with this device and a standard BP cuff used as the reference, so that the BP prediction fits the individual user. The same data from the ESP32 is also sent through the cloud to the alert component (Member 4).

Using these heart rate and BP values to check fall risk, the alert component sends the alert when needed. In this way, my component provides continuous, low-cost physiological monitoring that helps the complete system and the care staff to notice a possible fall or faint early.

1.3 Problem statement

Most fall detection systems use motion sensors, so they detect a fall only after it has happened. They can also give false alarms for normal activities, and they may miss faints, where the person slowly collapses. Many falls in older adults are linked to low blood pressure, abnormal heart rate, and fainting. Without monitoring these body signals, a fall detection system loses useful early information. There is a need for a low-cost wearable that monitors heart rate and blood pressure and gives a fall risk, so the full system can predict, confirm, and respond to falls better.

In hospitals and care homes, blood pressure (BP) is mostly checked by staff using a cuff, a few times a day. Continuous BP and heart rate monitoring is usually available only in intensive care units (ICU), which need expensive equipment. Between two manual checks, an elderly person's condition can become critical without anyone noticing, and a sudden drop in BP or an abnormal heart rate can lead to dizziness, fainting, or a fall. Existing motion-based fall detection systems cannot see these body changes. Also, fixed BP estimation models may not work well for every elderly person, because each person is different (for example: because of diseases, tissue thickness, or skin tone), and this can cause false alerts.

1.4 Research gap

Existing work mostly detects falls using motion sensors, and other work estimates BP from PPG signals. Continuous vital sign monitoring is mainly limited to ICU-level equipment, and manual cuff checks leave long gaps in monitoring. Many PPG-based BP models are trained on general datasets and do not consider individual differences, such as health conditions, tissue thickness, skin tone, or naturally abnormal BP and heart rate. This can cause wrong predictions and false alerts. Few studies combine a low-cost, easy-to-wear PPG device, a personalised BP model, and

fall risk prediction inside a complete fall detection system. [Update this after your literature review

1.5 Aim and objectives

Aim: To develop a low-cost, easy-to-wear PPG-based wearable component that continuously estimates heart rate and blood pressure using a personalised model, and predicts fall risk, in order to improve the accuracy and reliability of a complete elder-care fall detection system and to reduce the monitoring workload of care staff.

Objectives:

1. To evaluate how fall risk information from this component supports the overall fall detection decision (for example, by reducing false alarms and giving early warning).
2. To collect raw PPG signals using the ESP32 and MAX30102 and calculate heart rate and pulse intervals.
3. To build a personalized BP prediction model using a dataset collected with a standard BP cuff, So that individual differences cause fewer false alerts.
4. To train and evaluate a machine learning model that predicts systolic and diastolic BP.
5. To predict fall risk using heart rate and BP.
6. To send data in real time through MQTT (HiveMQ) to the backend and dashboard.
7. To test the system and integrate it with the other components.

1.6 Research questions or hypotheses

1. **RQ1:** How accurately can systolic and diastolic BP be predicted from MAX30102 PPG features?
2. **RQ2:** Can heart rate and predicted BP be used to classify fall risk reliably?
3. **RQ3:** Can the system send and show data in real time with low delay?

1.7 Scope limitations and assumptions

Scope: fingertip PPG using the MAX30102, BP prediction using machine learning, fall risk classification, MQTT data transfer, and dashboard display.

Limitations:

1. The dataset is small and comes from [number] participants.
2. The system is a research prototype and not a certified medical device.
3. Movement can add noise to the PPG signal.
4. The model may need re-calibration for each person.

Assumptions:

1. The sensor is worn correctly on the fingertip.
2. WiFi and internet are available.
3. The cuff readings are accurate enough to be used as the reference.

1.8 Individual contribution

My contributions are:

1. Building the ESP32 and MAX30102 hardware and firmware.
2. Peak detection and calculation of BPM and pulse intervals.
3. Collecting the personalised dataset with a BP cuff.
4. Training the BP prediction model.
5. Setting up the MQTT communication using HiveMQ and the backend subscriber.
6. Fall risk prediction

The other group members worked on the remaining components of the integrated system:

- **Member 1:** Motion-based fall detection using an accelerometer.
- **Member 2:** Environmental sensing to detect high-risk surrounding conditions.
- **Member 4:** Centralized alert and notification system.

1.9 Thesis structure

This thesis has eight chapters. Chapter 1 introduces the problem, aim, and objectives. Chapter 2 reviews related work. Chapter 3 explains the research methodology. Chapter 4 describes the design of the solution. Chapter 5 explains the implementation, integration, and testing. Chapter 6 presents the results. Chapter 7 discusses the findings and limitations. Chapter 8 gives the conclusion and future work.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This literature review adopts a structured approach to examine prior research, commercial technologies, and methodological frameworks relevant to fall detection, environmental hazard monitoring, physiological tracking, and automated alert routing for elderly care. The scope encompasses peer-reviewed studies published over the past decade, focusing specifically on Internet of Things (IoT) architectures, wearable inertial measurement units (IMUs), ambient environmental sensing, computer vision-based obstacle detection, photoplethysmography (PPG) signal processing, and context-aware decision support systems. Studies that address only isolated aspects—such as standalone wearable fall alarms without environmental or physiological context—are examined to identify limitations in current technological paradigms.

2.2 Domain and theoretical foundations

The theoretical foundation of this research rests on the multi-factorial nature of fall risk among older adults as defined by public health frameworks, combined with pervasive computing and ambient assisted living (AAL) principles. According to the World Health Organization, falls are rarely attributable to a single isolated trigger; rather, they emerge from the intersection of intrinsic physiological vulnerabilities (e.g., postural instability, sudden blood pressure drops, cardiovascular anomalies) and extrinsic environmental hazards (e.g., wet floors, inadequate illumination, physical obstacles). Pervasive computing theory supports the distribution of sensing nodes across the living environment and the human body to achieve continuous, non-intrusive monitoring. By integrating multi-modal telemetry streams through standardized messaging protocols (such as MQTT), the system bridges the theoretical gap between fragmented reactive monitoring and proactive, context-aware prevention

2.3 Related research

Extensive research has explored individual components of elder-care monitoring. In the domain of wearable inertial sensing, acceleration and angular rate data collected

from waist- or wrist-worn IMUs have been widely utilized to classify human activities and detect falls. While machine learning models trained on IMU data frequently report high laboratory classification accuracies, real-world deployments suffer from false positives triggered by rapid, non-fall movements (such as abruptly sitting on a couch). In environmental and vision-based monitoring, IoT ambient sensors and camera systems running object detection models (such as YOLO variants) have been deployed to detect home hazards. However, these vision systems often introduce significant privacy concerns and operate independently of biometric data streams. Furthermore, continuous physiological monitoring through photoplethysmography (PPG) has gained traction for tracking heart rate and estimating blood pressure, yet most biometric algorithms experience performance degradation during active physical movement due to motion artifacts.

2.4 Existing systems products and techniques

Commercially available elder-care technologies predominantly adopt reactive, single-modality approaches. Consumer medical alert pendants and wristbands typically rely on manual trigger buttons or basic threshold-based fall alarms that automatically route every detected incident to a centralized human call centre or emergency contact. Similarly, commercial smart home kits offer standalone environmental hazard detectors (such as smoke or flood alarms) or security cameras, but these systems do not correlate environmental risk levels with the resident's instantaneous physical activity or physiological state. Consequently, existing commercial products frequently suffer from high false-positive rates, lack flexible multi-tiered notification routing, and contribute to caregiver alert fatigue without addressing root environmental or physiological risk factors.

2.5 Gap synthesis and conceptual framework

The synthesis of existing literature and commercial products reveals a critical overarching research gap: the absence of a continuous, integrated, low-cost multimodal framework that simultaneously evaluates an older adult’s movement stability, environmental surroundings, and physiological state to execute context-aware, multi-tiered alert routing. Existing literature treats motion

recognition, hazard mapping, biometric tracking, and emergency response as disconnected silos.

To bridge this gap, the conceptual framework of this project unites four interdependent components—Context-Aware Activity Recognition, Environmental Risk Mapping, BioSignal Prediction, and Alert Classification—into a cohesive pipeline connected via an MQTT messaging broker. This unified framework ensures that fall events are validated against environmental contexts and routed intelligently to the appropriate responder (voice interface, family caregiver, or emergency services), thereby shifting elder-care technology from isolated reactivity to comprehensive, proactive prevention.

3 RESEARCH METHODOLOGY

3.1 Research design

This research uses an experimental and quantitative design. A wearable prototype was built and used to collect PPG signals together with reference BP readings. A machine learning model was then trained and tested to predict BP, and the predicted values with heart rate were used to estimate fall risk. The research has the following stages:

1. Hardware and firmware development (ESP32 and MAX30102).
2. Personalised data collection with a standard BP cuff as the reference.
3. Data preprocessing and feature extraction.
4. Model training and evaluation for BP prediction and fall risk prediction.
5. Real-time deployment through MQTT, the backend, and the dashboard.
6. System testing and integration with the other components.

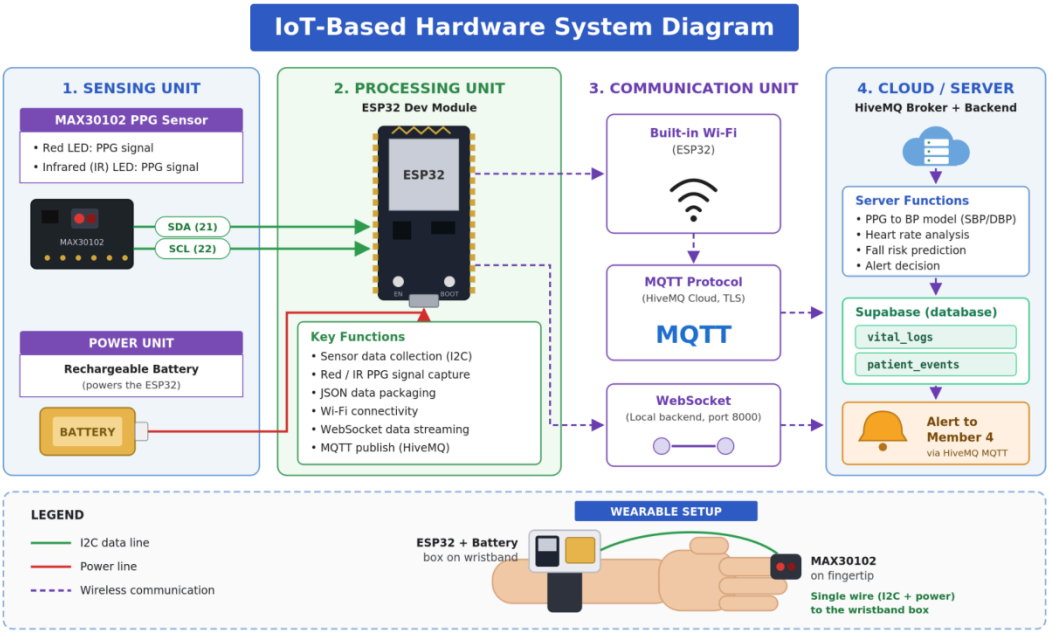


Figure 1: Overall System Architecture

3.2 Data participants or study materials

Participant. The dataset is personalised and was collected from one participant. The participant gave consent for the use of the data. Because only one person is used, the model is trained for this person and the results may not apply to other people (see Section 7.5).

Materials.

1. ESP32 microcontroller
2. MAX30102 PPG sensor (red and infrared LEDs)
3. Standard BP cuff monitor: used as the reference
4. Laptop with PlatformIO/VS Code for firmware upload and data logging
5. HiveMQ Cloud broker, Python backend, and database

Dataset. The dataset has [number] samples from [number] sessions. Each record has the following columns:

Column	Description
Timestamp_ms	Time of the sample in milliseconds
Red_Raw	Raw red-light PPG value
IR_Raw	Raw infrared PPG value
BPM	Heart rate in beats per minute
BeatAvg	Average heart rate over recent beats
PulseInterval_ms	Time between two Sequential heartbeats
Systolic_BP	Reference systolic BP from the cuff (mmHg)
Diastolic_BP	Reference diastolic BP from the cuff (mmHg)
Activity	Activity during recording (e.g. [resting, walking])

Figure 2:Dataset columns.

Dataset columns.

The dataset has no fall risk label, because falls and faints were not recorded. Fall risk is calculated from heart rate and BP using rules (see Section 3.5).

3.3 Experimental or study procedure

The data was collected by the author using the prototype device and a standard cuff-type BP monitor . The procedure was as follows:

1. The author sat comfortably and rested for 3 to 5 minutes before each recording session to get a stable baseline.
2. The MAX30102 sensor was placed on the finger and fixed firmly with a strap to reduce noise. Finger movement and changes in pressure can add noise to PPG signals, so the sensor was held tightly and securely. However, it was not fastened too tightly or too loosely, because both cases gave unstable or wrong signal values.
3. The ESP32 recorded raw Red and IR PPG signals at a sampling rate of 50 Hz. It calculated BPM, the average BPM (BeatAvg), and pulse intervals using peak detection, and logged each sample with a timestamp in milliseconds.
4. At the same time, a BP reading was taken with the standard cuff on the arm. The cuff takes about one minute to give one reading, so only one systolic and one diastolic value was available for each measurement. the PPG recording started when the cuff started, and ended when the cuff showed the result.
5. The cuff readings were then **added manually** to the dataset as the Systolic_BP and Diastolic_BP columns. Each cuff reading was assigned to all PPG samples recorded during the matching measurement period.
6. Steps 2 to 5 were repeated for about 5 to 7 readings per session over 15 sessions and about 20 days. Different activities and conditions (for example: resting, after walking, after climbing stairs) were used to get a range of BP and heart rate values. The activity was recorded in the Activity column.
7. Each record in the final dataset therefore contains the raw PPG values, the heart rate values from the ESP32, the reference BP from the cuff, and the activity.

3.4 Measures and evaluation metrics

BP prediction (regression: Random forest). The model predicts systolic and diastolic BP. It is evaluated with:

1. Mean Absolute Error (MAE) in mmHg.
2. Coefficient of determination (R^2).
3. The percentage of predictions within ±5 mmHg and within ±10 mmHg of the cuff reading.
4. Mean error (bias) and standard deviation (SD) of the error, compared with the AAMI/ISO 81060-2 limits (mean error within ±5 mmHg and SD within 8 mmHg).
5. Accuracy, calculated as 100 − MAPE (mean absolute percentage error), is also reported only as an easy-to-read extra value. It is not a standard BP validation measure.

All results are compared with a **baseline model that always predicts the mean BP**. The model is useful only if it performs better than this baseline.

Fall risk (rule-based). The dataset has no fall risk label. The risk rules are therefore evaluated in this way: the reference risk is calculated by applying the rules to the cuff BP and measured heart rate, and the predicted risk is calculated by applying the same rules to the predicted BP and heart rate. The two are compared using accuracy, precision, recall, F1-score, and the confusion matrix. Recall for high-risk cases is the most important value, because a missed high-risk case is worse than a false alarm. This evaluates how well the risk level is kept. It does not show that real falls are predicted.

System performance.

- End-to-end delay from the ESP32 to the dashboard.
- MQTT message delivery rate.
- BPM accuracy compared with a reference device. (Standard BP cuff)

3.5 Data analysis

Data preparation. The dataset has 23,930 samples recorded at about 50 Hz. Column names were cleaned, the rows were sorted by timestamp, and duplicate and missing values were checked. No duplicate rows, no duplicate timestamps, and no missing values were found.

Features and targets. The input features are Red_Raw, IR_Raw, BPM, and BeatAvg. The targets are Systolic_BP and Diastolic_BP. The PulseInterval_ms column was not used, because in this dataset it is a counter of the time since the last detected beat and not a true beat-to-beat interval.

Model. A Random Forest regressor (100 trees, minimum 10 samples per leaf, random seed 42) was used to predict both BP values together.

Validation. Samples are only 20 ms apart, so neighbouring rows are almost the same. A random split would put almost identical rows in both the training and testing sets and give a wrongly high score. Therefore, two group-based validations were used:

1. ***Time-block cross-validation****:* the data was divided into 48 blocks of 10 seconds, and 5-fold group cross-validation was used so that a block is never in both training and testing.
2. ***Leave-one-activity-segment-out****:* the data has 5 activity segments, and the model was tested on one segment while trained on the others. This is a hard test, because the model is tested on an activity it has not seen.

Both validations were compared with the mean-prediction baseline. Feature importance was calculated from the final model. The final model used by the backend was trained on all data and saved as `bp_model.joblib`, which includes the feature list.

Fall risk. Fall risk is decided by threshold rules using systolic BP, diastolic BP, and heart rate, with thresholds from clinical guidelines for hypotension and heart rate range.

Tools. Python, pandas, NumPy, scikit-learn, and joblib.

Validation	Model	Systolic MAE	Diastolic MAE	R^2
Time-block (10 s)	Baseline	11.65	7.25	-0.001
Time-block (10 s)	Random Forest	4.91	3.46	0.763
Unseen activity	Baseline	16.82	10.14	-0.683
Unseen activity	Random Forest	12.29	7.34	0.026

Figure 3:Data Analysis

The model performs well when it has seen similar conditions (time-block test) but its performance drops when tested on an activity it has not seen. Feature importance shows that the model mostly depends on heart rate (BeatAvg, 79%), and the raw PPG values contribute little. This means that, with this small one-person dataset, the model captures the link between heart rate and BP and not the PPG waveform itself. The results are therefore a proof of concept and not a validated BP measurement.

3.6 Ethics privacy security and responsible AI

Human data and consent. The only human data used in this research is the author's own heart rate, PPG signals, and BP readings. The author collected these data from herself and gave consent for their use in this study. No other person took part, and no data was collected from patients or elderly people.

Safety of the data collection. The measurements were non-invasive. The MAX30102 sensor uses low-power red and infrared LEDs, and a commercial BP cuff was used in the normal way. The sensor was fixed firmly on the fingertip, but not so tightly that it blocked blood flow, and recording was stopped if there was any discomfort.

Privacy. The dataset has no name, ID number, or contact details. It contains only sensor values, BP values, and the activity type. The dataset and the code are stored in a private repository , and the dataset will be shared only with the supervisors and examiners. Health data is personal even for one person, so the dataset will not be published publicly without removing or checking the content first.

Security of the system.

- *Data in transit:* the ESP32 and the Python backend connect to the HiveMQ Cloud broker over TLS, so the data is encrypted between the device, the broker, and the backend.
- *Authentication:* the broker needs a username and password, and access is limited to the project's clients.
- *Credentials:* broker passwords and other secrets are kept in environment variables , and are not written in the thesis or in public code.
- *Topic design:* data is sent on a dedicated topic (`patient/vitals`), and the backend subscribes only to this topic.

Known limits. This system is a research prototype, and the following limits apply:

- *Not tested on elderly users:* the system was tested only by the author. It has not been used or tested with any elderly person, so its comfort, ease of wearing, and signal quality for older users are not known.
- *Not tested in a real care setting:* the dashboard and the data flow work in the lab test, but they have not been tried with real care staff or in a care home or hospital.
- *Not clinically validated:* the BP values and the fall risk are not checked by medical professionals or against clinical standards.
- *No user login on the dashboard:* anyone who has the dashboard link can see the data.
- *Single device only:* the system was tested with one ESP32 only.

3.7 Reproducibility and research artefacts

Artefact	Description	Location
Firmware	ESP32 code (PlatformIO) for reading the MAX30102, peak detection, BPM, and MQTT publishing	HirushiDivya/research-c-v2
Backend	Python backend: MQTT subscriber, BP prediction, database storage, WebSocket to dashboard	HirushiDivya/research-backendv2
Training script	`train_model.py`: data cleaning, validation, model training	HirushiDivya/research-backendv2
Dataset	`collected_data.csv`: 23,930 samples, 9 columns	HirushiDivya/research-backendv2
Trained model	`bp_model.joblib`: Random Forest model with its feature list and targets	HirushiDivya/research-c-v2
Dashboard	React dashboard	HirushiDivya/research-frontendv2
Hardware	Wiring diagram and component list	Appendix A

Figure 4:Reproducibility and research artefacts

Software and hardware versions. [Fill in your real versions.]

- Python [3.14.8], pandas [3.0.6], NumPy [2.5.3], scikit-learn [1.9.1], joblib [version],scipy [1.18.1]
- Backend framework: FastAPI [0.142.2], MQTT library: [paho-mqtt 2.1.0]

- Firmware: PlatformIO with esp32dev , MAX30102 [SparkFun MAX3010x Pulse and Proximity Sensor Library v1.1.2]
- Dashboard: React [18.2.0]
- MQTT broker: HiveMQ Cloud (free tier)[PubSubClient v2.8]

Experiment settings.

- Sampling rate: about 50 Hz (about 20 ms between samples).
- Model features: Red_Raw, IR_Raw, BPM, BeatAvg. Targets: Systolic_BP and Diastolic_BP.
- Model: Random Forest Regressor, 100 trees, minimum 10 samples per leaf, random seed 42.
- Validation: 5-fold group cross-validation on 48 time blocks of 10 s (500 rows each), and leave-one-activity-segment-out on 5 activity segments.
- MQTT topic: `patient/vitals`. The JSON message format is shown in Chapter 4.

How to reproduce the results.

1. Place `collected_data.csv` in the same folder as the training script.
2. Install the packages from `requirements.txt` in a virtual environment.
3. Run `python train_model.py`. It prints the validation results and saves `bp_model.joblib`.
4. Start the backend, which loads the model and subscribes to the broker.
5. Upload the firmware to the ESP32 with PlatformIO, and set the WiFi and broker settings .

Limits of reproducibility.

- The dataset depends on one person, one device, and one sensor placement. Another person who repeats the work will collect different data, so the numbers will differ.

- The cuff readings were added by hand, and the timing match with the PPG recording depends on how it was done (see Section 3.3).
- Random Forest results depend on the random seed and the library version. The seed is fixed to reduce this.
- Broker credentials are not included in the shared code. A new user needs to create their own HiveMQ account and set their own credentials.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Functional requirements

ID	Requirement
FR1	The device shall read raw Red and IR PPG signals from the MAX30102 at about 50 Hz.
FR2	The device shall detect heartbeat peaks and calculate BPM and the average BPM (BeatAvg).
FR3	The device shall send the data in JSON format to the cloud broker using MQTT.
FR4	The backend shall receive the data from the broker as soon as it is published.
FR5	The backend shall predict systolic and diastolic BP using the trained model.
FR6	The backend shall save the data and predictions in the database.
FR7	The backend shall send the live data to the dashboard in real time.
FR8	The dashboard shall show heart rate, predicted BP, and fall risk status.
FR9	The data shall be available to the alert component (Member 4) so that it can send alerts.
FR10	The system shall be able to support more than one ESP32 device without changing the backend.

Figure 5:Functional Requirements

Non-functional requirements

ID	Requirement
NFR1	*Low cost:* the device shall use cheap, easily available parts.
NFR2	*Comfort:* the device shall be easy to wear (fingertip sensor, small, light).
NFR3	*Real time:* the delay from the sensor to the dashboard shall be [e.g. below 2 seconds].
NFR4	*Security:* data shall be encrypted in transit and the broker shall need authentication.
NFR5	*Reliability:* the device shall reconnect automatically if WiFi or the broker connection is lost.
NFR6	*Accuracy:* the BP model shall perform better than the mean-prediction baseline, and it is compared with the AAMI limits.
NFR7	*Scalability:* new devices shall be added by connecting them to the broker only.
NFR8	*Privacy:* no personal identity data shall be stored in the sensor data.

Figure 6:non funtional requirements

4.2 Component architecture within the complete solution

The complete system is an elder-care fall detection system with four parts:

1. A motion-based fall detection component that uses accelerometer data (x, y).
2. An environmental sensing component.
3. An alert component (Member 4).
4. **My component:** the PPG-Based Biometric Fall Risk Prediction System.

My component is the physiological layer. It watches heart rate and BP, which can show early signs of a faint or a fall, while the motion component detects the fall itself.

My component has five layers:

Layer	Parts	Role
Sensing	MAX30102	Measures Red and IR light absorption in the fingertip
Edge processing	ESP32	Reads signals, detects peaks, calculates BPM, builds JSON
Communication	HiveMQ Cloud (MQTT over TLS)	Receives data from the ESP32 and passes it to subscribers
Backend	Python backend, ML model, database	Receives data, predicts BP, saves data, pushes live data
Application	React dashboard, alert component (Member 4)	Shows data to users and sends alerts

Figure 7:Layers of component

Data flow

1. The ESP32 publishes JSON data to the topic `patient/vitals` on HiveMQ.
2. The HiveMQ broker passes the message to all subscribers.
3. The Python backend (a subscriber) receives it immediately, predicts BP, and saves the record in supabase.
4. The backend sends the live data to the React dashboard through WebSockets.
5. The alert component (Member 4) also receives the data through the cloud, and sends the alert when needed.

Interfaces with the other components. My component and the alert component share the MQTT topic and the JSON format (Section 4.4). [I applied the fall risk rules: my backend]

4.3 Design alternatives and decisions

Decision 1: Communication between the ESP32 and the backend

Option	Advantages	Disadvantages
Direct connection (ESP32 to Python backend)	Simple, no extra service	Both must be on the same WiFi network; the laptop IP address changes after a restart; each new device needs backend changes; the backend must always be running
MQTT broker (publish/subscribe)	Devices and backend are separated; more devices can be added by connecting to the broker only; works over the internet	Needs a broker and internet

Figure 8: Communication alternatives between the ESP32 and the backend

Decision: MQTT broker. It solves the same-network and changing-IP problems and makes it easy to add more ESP32 devices, which is needed for elder care with many users. It useful when send real time data to member 3.

Decision 2: Broker platform

Option	Advantages	Disadvantages
HiveMQ Cloud (free tier)	Easy cloud setup; built-in WebSocket support; TLS/SSL security by default	Free tier limits [state limits]
Self-hosted broker (e.g. Mosquitto)	Full control, no usage limits	Needs a server, setup, and security configuration
Other cloud brokers (e.g. AWS IoT Core / EMQX Cloud)	Scalable cloud infrastructure; advanced security and device management	Complex setup process; potential cost implications beyond free tiers

Figure 9:alternatives for broker platform

Decision: HiveMQ Cloud, because of easy deployment, built-in WebSockets, and TLS out of the box.

Decision 3: Protocol

Option	Notes
MQTT	Light, designed for small IoT devices, publish/subscribe, low bandwidth
HTTP (REST)	Needs the device to send a request each time; heavier; no push to subscribers
Direct WebSocket from the ESP32	Needs a fixed server address and the same network; one-to-one

Figure 10:alternatives for protocol

Decision: MQTT. MQTT was adopted after initially prototyping with direct WebSockets, chosen for its lightweight publish-subscribe overhead and superior reliability for continuous IoT telemetry.

Decision 4: BP prediction model

Option	Notes
Linear regression	Simple, but may not capture non-linear relations
Random Forest	Handles non-linear relations, works well on small tabular data, little tuning, gives feature importance
Neural network	Needs much more data than the data available; more risk of overfitting

Figure 11:BP prediction Model

Decision: Random Forest, because the dataset is smal.

Decision 5: Fall risk method

Option	Notes
Machine learning classifier	Needs labelled falls or faints, which were not collected
Threshold rules on BP and heart rate	Simple, easy to explain, based on clinical ranges, does not need labels

Figure 12:alternatives for fall risk prediction

Decision: threshold rules, because there is no fall risk label in the dataset and the rules can be explained to care staff. Furthermore, collecting real fall data from elderly subjects is ethically unfeasible and unsafe.

Decision 6: Sensor position and input features

Option	Notes
Fingertip	Strong PPG signal, but less comfortable for long use
Wrist	More comfortable, but weaker signal and more motion noise

Figure 13:alternatives for senser position

Decision: fingertip, held firmly, because the signal is stronger and the noise is lower. [Future work: wrist version.] The model uses Red_Raw, IR_Raw, BPM, and BeatAvg.

Decision 7: Dashboard updates. WebSockets were chosen so that the backend can push data to the dashboard immediately, without the dashboard asking repeatedly

4.4 Detailed design

4.4.1 Hardware design

- ESP32 connected to the MAX30102 over I2C (SDA, SCL, VIN, GND).
- Power: battery.
- The sensor is fixed on the fingertip with strap

Figure 4.3: Wiring diagram.

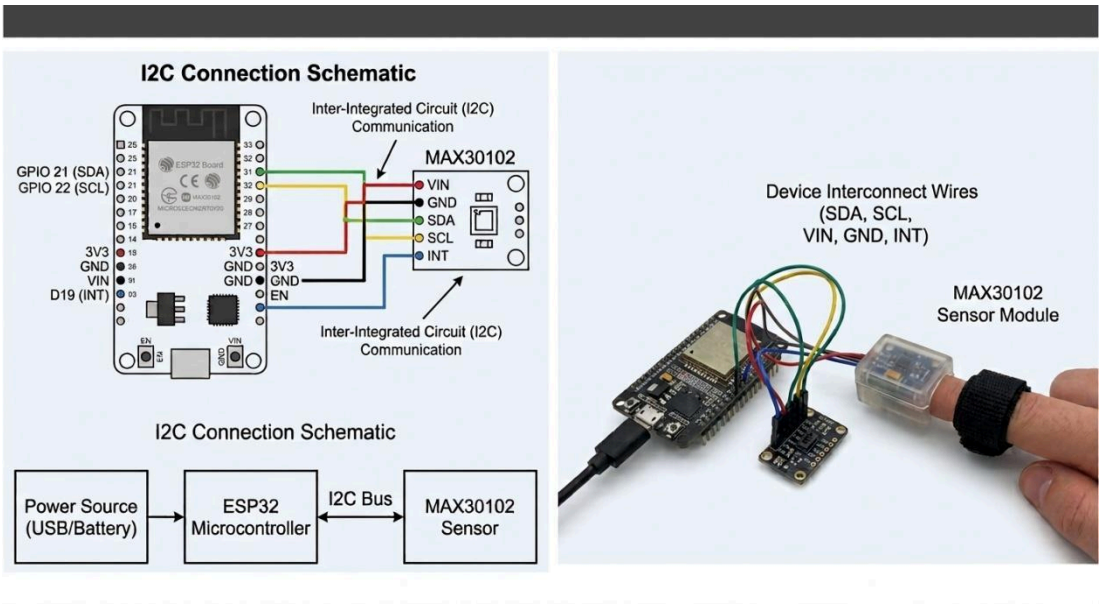


Figure 4.4: Photo of the device.

4.4.2 Firmware design (ESP32)

1. Connect to WiFi, then connect to HiveMQ over TLS with the username and password.
2. Read Red and IR values from the MAX30102 at about 50 Hz.
3. Detect a heartbeat peak in the IR signal.
4. Calculate BPM from the time between peaks, and calculate BeatAvg as the average of the last 5second readings.

5. Check that a finger is present, and skip invalid readings.
6. Build the JSON message and publish it to `patient/vitals`.
7. If the connection is lost, reconnect automatically.

Figure 4.5: Firmware flowchart.

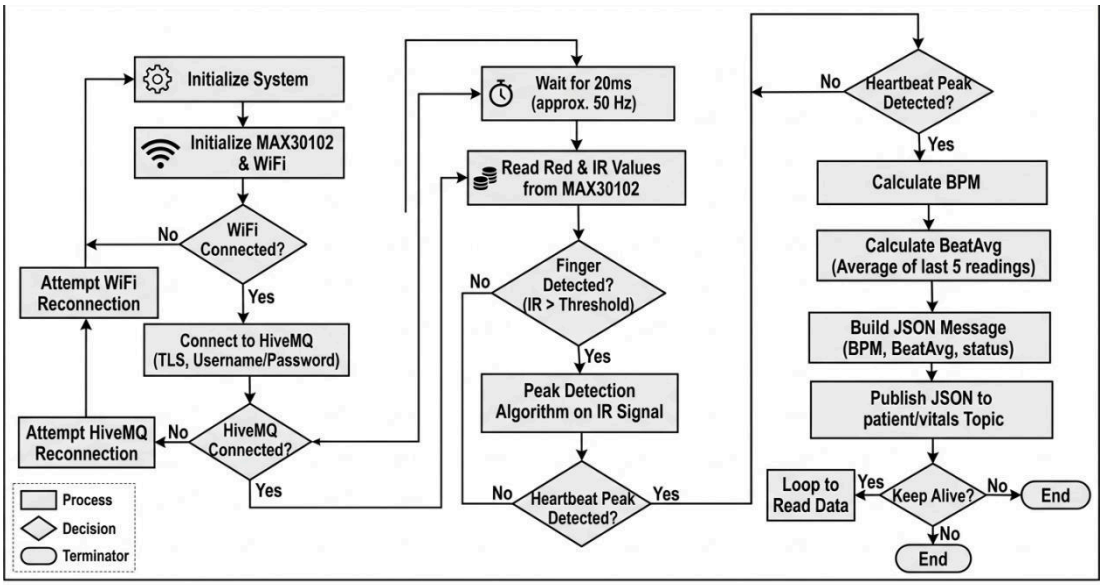


Figure 4.5: Firmware Flowchart for ESP32 and MAX30102 Vitals Monitoring System

4.4.3 Message design (JSON)

Topic: `patient/vitals`

```
{
 "timestamp_ms": 123456,
 "red_raw": 51234,
 "ir_raw": 60234,
 "bpm": 72,
 "beat_avg": 70
}
```

4.4.4 Backend design

1. Subscribe to `patient/vitals` on HiveMQ.

2. Validate each message (required fields, valid ranges).
3. Load the model from `bp_model.joblib` once at startup.
4. Predict systolic and diastolic BP from Red_Raw, IR_Raw, BPM, and BeatAvg.
5. Calculate the fall risk status using the threshold rules (Section 4.4.6).
6. Save the record in the database.
7. Send the result to the dashboard through WebSockets.

Figure 4.6: Backend sequence diagram.

4.4.5 Database design

Table 1: `vital_logs` Table Design

This table shows storing data like blood pressure and heart rate predicted by ML from the ESP32 sensor every 5 seconds.

Field	Type	Description
Id	int(8) primary key	Unique identifier for each vital log record
created_at	timestamptz	Timestamp when the vital record was inserted
heart_rate	float4	Averaged heart rate (BeatAvg)
systolic_bp	int4	Predicted systolic blood pressure
diastolic_bp	int4	Predicted diastolic blood pressure
pulse_interval	float4	Pulse timing interval (pti)
patient_status	text	Calculated health risk status (e.g., Normal, Risky)

Table 2: `patient_events` Table Design

This shows storing the patient's daily activities (Patient Activities/Events - e.g., Medication, Exercise, Sleep, Meditation) and notes.

Field	Type	Description
id	`int8` (Primary Key)	Unique identifier for each event log
event_time	timestamptz	Timestamp when the event or activity occurred
activity_type	text	Type of activity (e.g., Medication, Exercise, Sleep)
notes	text	Additional notes or descriptions regarding the event

4.4.6 Fall risk rules

Condition	Risk status
140 mmHg < 1-min average systolic BP < 90 mmHg	High (code: Critical)
45 bpm >1-min average heart rate > 120 bpm	High (code: Critical)
1-min average systolic BP 90-99 mmHg or 131-140 mmHg	Medium (code: Risky)
1-min average heart rate 45-49 bpm or 101-120 bpm	Medium (code: Risky)
Otherwise (systolic 100-130 mmHg and heart rate 50-100 bpm)	Low (code: Normal)

4.4.7 Dashboard design. The React dashboard receives the live data through WebSockets and shows heart rate, predicted systolic and diastolic BP, the fall risk status, and a graph of recent values. [Figure 4.7: Dashboard screenshot.]

4.5 Standards and professional practice

Area	Standard or practice	How it is used
Messaging	MQTT (OASIS standard)	Publish/subscribe data transfer
Data format	JSON (RFC 8259)	Message format
Security	TLS	Encrypted connection to HiveMQ
BP accuracy	AAMI/ISO 81060-2	Used as a reference for error limits
Wireless	IEEE 802.11 (WiFi), I2C bus	ESP32 connection and sensor connection
Software practice	Version control (Git), modular code, saved model with feature list, testing	Maintainability and reproducibility
Medical	The system is not a certified medical device (no clinical approval)	Stated clearly in Chapter 3.6
Ethics	Informed consent, privacy of health data	Section 3.6
Safety	Low-power LEDs, non-invasive sensing	Section 3.6

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

Layer	Tool / technology	Purpose
Hardware	ESP32 (esp32dev board), MAX30102 sensor, battery	Sensing and data sending
Firmware	PlatformIO in VS Code, C++ (Arduino framework), [MAX30102 library], [MQTT library]	Reading the sensor, peak detection, MQTT publishing
Communication	HiveMQ Cloud (MQTT over TLS)	Cloud broker
Backend	Python, FastAPI, paho-mqtt	Receiving data, BP prediction, saving, WebSocket
Machine learning	Python, pandas, NumPy, scikit-learn, joblib	Model training and saving
Database	Supabase (tables `vital_logs` and `patient_events`)	Storing the records
Frontend	React	Live dashboard
Tools	Git/GitHub, VS Code, Serial Monitor, MQTT client tool(HiveMQ Web Client)	Version control, debugging

5.2 Component implementation

5.2.1 Hardware assembly. The MAX30102 sensor is worn on the fingertip and is connected by one wire to a small box that holds the ESP32 and the battery. The box is fixed on a wristband. The sensor connects to the ESP32 over I2C .



Figure 5.1: Photo of the device.

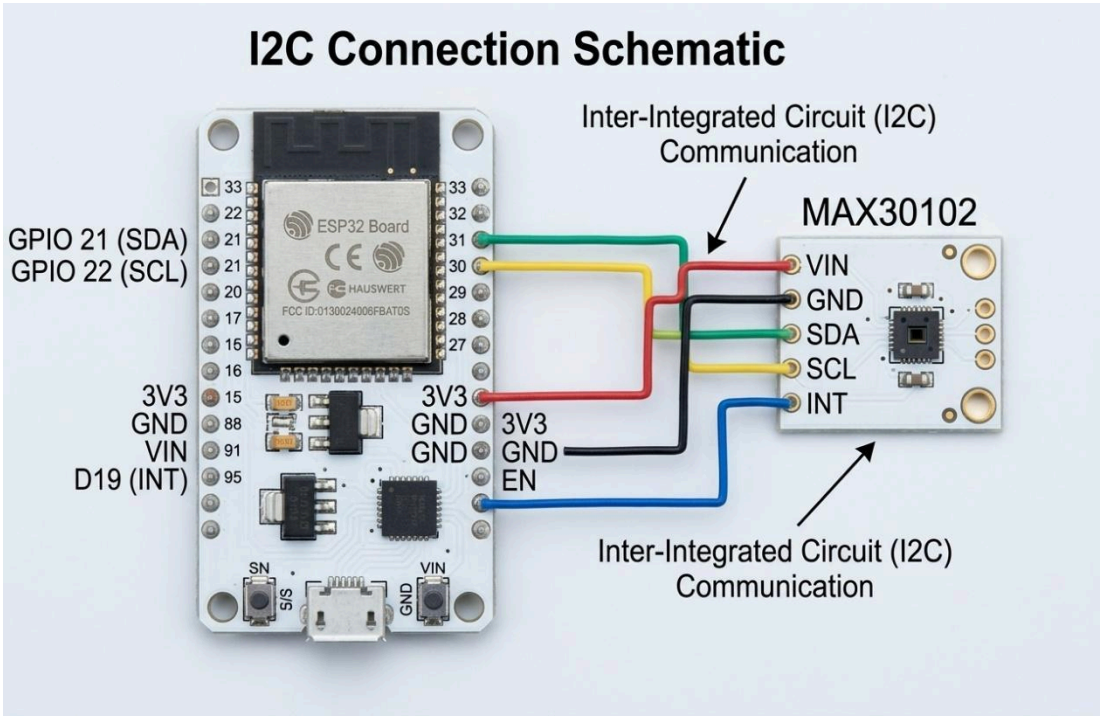


Figure 5.2: Wiring diagram.

5.2.2 Firmware. The firmware was written in PlatformIO. It does the following:

1. Connects to WiFi and then to the HiveMQ broker using TLS, a username, and a password.
2. Reads Red and IR values from the MAX30102 at about 50 Hz.
3. Checks that a finger is on the sensor, and ignores invalid readings.
4. Detects heartbeat peaks in the IR signal, and calculates BPM and BeatAvg.
5. Builds a JSON message and publishes it to `patient/vitals`.
6. Reconnects to to the broker automatically if the connection is lost.

5.2.3 Data collection and model training. The same firmware was used in a logging mode to collect the dataset (Section 3.3). The training script `train_model.py` cleans the data, runs the group-based validation, trains the final Random Forest model with the features Red_Raw, IR_Raw, BPM, and BeatAvg, and saves it as `bp_model.joblib` together with the feature list.

5.2.4 Backend. The FastAPI backend does the following:

1. Loads `bp_model.joblib` once at startup.
2. Subscribes to the topic `patient/vitals` on HiveMQ.
3. Validates each incoming message.
4. Predicts systolic and diastolic BP.
5. Calculates the fall risk status using the threshold rules.
6. Saves the record in Supabase (table `vital_logs`). and handles patient activity logs and annotations (`patient_events`) via dedicated endpoints.
7. Sends the result and health status to the connected dashboards through WebSockets.

The first version used a direct WiFi WebSocket connection between the ESP32 and the local backend, and that it was changed to MQTT through HiveMQ because of the problems explained in Section 4.3.

5.2.5 Dashboard. The React dashboard connects to the backend through WebSockets and shows the live heart rate, predicted BP, fall risk status, graph of recent values and Patient Events .

5.3 Integration with the other components

Alert component (Member 4): the ESP32 data goes through the HiveMQ broker, and the alert component also subscribes to the topic `patient/vitals`. Raw sensor data directly streamed from the device is handled by Member 4 .

Deployment of the backend and frontend:[the backend and frontend run on a team member's laptop, and the ESP32 sends data to the broker, so the device does not need to be on the same network as the backend.

Problems found during integration and how they were solved:

1. Data routing and subscription mismatch across team modules: During initial integration, sensor telemetry successfully reached the HiveMQ broker and backend, but data packets were not properly routed to Member 4's module. This was resolved by verifying MQTT topic naming conventions (`patient/vitals`), ensuring correct JSON payload formatting, and confirming that all subscribing clients shared the exact broker credentials and topic subscriptions.

5.4 Verification and software testing

Test approach. The component was tested at four levels: unit tests, component tests, integration tests, and system tests.

ID	Level	Test	Expected result	Actual result	Status

T1	Unit	Peak detection with a finger on the sensor	A stable BPM value is calculated	BPM stable between 60 and 100 after 2 min, compared with [reference device] reading of [value].	Pass
T2	Unit	No finger on the sensor	No valid reading is sent	The dashboard display zero values.	Pass
T3	Unit	JSON message format	All required fields are present	All fields present.	Pass
T4	Component	ESP32 publishes to HiveMQ	The message appears in the broker client	Message appeared within 1min, about 50 messages per second	Pass
T5	Component	Backend receives a message	The message is received and decoded	Backend printed the values coming from esp32.	Pass

T6	Component	Model prediction from a sample input	Systolic and diastolic values are returned	Returned systolic 113mmHg and diastolic 72mmHg mmHg for the input [102412,31268,36225,99.67,92,561,113,72,resting,Normal]	Pass
T7	Component	Invalid or missing fields	The message is rejected without a crash	Backend ignored the message	Pass
T8	Integration	ESP32 → HiveMQ → backend → Supabase	A new record appears in `vital_logs`	New rows with correct 12 fields appeared, rows in 1 min.	Pass
T9	Integration	HiveMQ → Member 4	The alert component receives the data	Dashboard updated about every 1s	Pass

T10	Integration	WiFi lost and restored	The device reconnects and sends data again	Member 4 received the data on the same topic, confirmed by same data display both my and component 4 dashboard.	Pass
T11	System	WiFi lost and restored	The device reconnects and sends data again	Device reconnecte d after about 1min, and publishing continued	Pass
T12	System	Wrong broker credentials	The connection is refused	Connection refused, Serial Monitor showed error message.	Pass
T13	System	Normal use for more than 20 minutes	No crash and no data loss	messages sent 3,000 , received 2540, no crash.” Delivery rate =84.67%	Pass

Performance checks.

- End-to-end delay (ESP32 to dashboard)
- Message delivery rate.
- BPM accuracy compared with reference device..

5.5 Deployment and operation

Device: the ESP32, the battery, and the sensor are placed in a box on a wristband, and the sensor is fixed on the fingertip. Battery life: 3 hours

.**Cloud:** HiveMQ Cloud free with 2 users, TLS enabled, and the topic `patient/vitals`.

Backend and dashboard: run on a team member's laptop with [how to start them. `uvicorn main:app`, and they need internet access.

Database: Supabase project with the tables `vital_logs` and `patient_events`.

Configuration: WiFi name and password, broker address, username, and password are set in the config file , and are not stored in the public code.

6 RESULTS AND VALIDATION

6.1 Research results

6.1.1 Dataset summary. The dataset has 23,930 samples recorded at about 50 Hz, which is about 8 minutes of data from one participant (the author). It has no missing values, no duplicate rows, and no duplicate timestamps. The Activity column has resting (11,930 samples) and walking, running, tired, and sleeping (about 3,000 samples each). The reference BP has 58 unique systolic values and 41 unique diastolic values, so the number of independent BP values is small .

6.1.2 BP prediction results. The Random Forest model was compared with a baseline that always predicts the mean BP. Two group-based validations were used (Section 3.5).

Validation	Model	Target	MAE (mmHg)	Within ±5 mmHg	Within ±10 mmHg	R^2
Time-block (5-fold, 10 s blocks)	Baseline	Systolic	11.65	19.5%	48.7%	-0.001
		Diastolic	7.25	42.0%	74.1%	
	Random Forest	Systolic	4.91	60.7%	87.3%	0.763
		Diastolic	3.46	73.3%	97.6%	
Unseen activity (leave-one-segment-out)	Baseline	Systolic	16.82	6.0%	16.2%	-0.683
		Diastolic	10.14	12.5%	51.5%	
	Random Forest	Systolic	12.29	19.5%	37.6%	0.026
		Diastolic	7.34	32.0%	71.0%	

R^2 is shown once for both targets together, as calculated by the training script.

Reading the results.

- When the model has seen similar conditions (time-block test), it is clearly better than the baseline. The systolic MAE is 4.91 mmHg compared with 11.65 mmHg, and the diastolic MAE is 3.46 mmHg compared with 7.25 mmHg.
- When the model is tested on an activity it has not seen, the performance drops a lot. The systolic MAE is 12.29 mmHg and R^2 is 0.026, which is only a small improvement over the baseline (16.82 mmHg).

- The time-block result is probably still too optimistic. The blocks come from the same short recording, so neighbouring blocks have similar BP values.

AAMI comparison

The mean error and the standard deviation (SD) of the error were compared with the AAMI limits (mean error within ±5 mmHg and SD within 8 mmHg). In the time-block test, the Random Forest model was inside the limits for both targets (systolic: mean error 0.67 mmHg, SD 6.37 mmHg; diastolic: mean error 0.38 mmHg, SD 4.48 mmHg), while the baseline model was outside the limits. In the unseen-activity test, the model was outside the limits for systolic BP (mean error 6.13 mmHg, SD 12.65 mmHg) and inside the limits for diastolic BP (mean error 3.79 mmHg, SD 7.56 mmHg), but the diastolic SD is very close to the limit. This comparison is only a guide. The AAMI/ISO 81060-2 protocol needs many participants and a specific measurement procedure, and this study used one participant, so it does not claim to follow the full protocol.

Validation	Model	Target	Mean error (mmHg)	SD of error (mmHg)	Within AAMI limits?
Time-block	Baseline	Systolic	-0.00	13.81	No
		Diastolic	-0.00	8.87	No
	Random Forest	Systolic	0.67	6.37	Yes
		Diastolic	0.38	4.48	Yes
Unseen activity	Baseline	Systolic	3.65	17.96	No
	Diastolic	Diastolic	1.99	11.05	No
Random Forest	Systolic	Systolic	6.13	12.65	No
	Diastolic	Diastolic	3.79	7.56	Yes

6.2 Product and technical results

Measure	Result	How it was measured
End-to-end delay (ESP32 to dashboard)	Mean 0.8 s, min 0.5 s, max 1.2 s, n = 10	When finger take from sensor all values display as 0.
Message delivery rate	messages sent 3,000 , received 2540, no crash." Delivery rate =84.67%	Keep finger1 min and calculate data receive to backend.
BPM accuracy compared with Standard BP cuff device	Display same range values both	Same time using standard BP cuff and my device get BP values.
Battery life	About 3 hours full charge to shutdown	Battery full charge and used until shutdown.
Dashboard update interval	About 20 updates per second	Check using /ws/client
Cost of the device	About 6000 - 8000	ESP32, MAX30102, battery, charger module, box, wristband, cover

6.3 User or stakeholder validation

The system was used and tested only by the author. It has not been tested with elderly people, care staff, or patients.

6.4 Integration results

- **Device to cloud:** the ESP32 sends data to HiveMQ over TLS, and the backend receives it.
- **Backend to database and dashboard:** records are saved in `vital_logs`, and the live data is shown on the React dashboard through WebSockets
- **Alert component (Member 4):** Member 4 received the data
- **Problems found and solved:**
    1. changing from direct WebSocket to MQTT: Direct WebSocket connections between the ESP32 and the backend caused instability and connection drops under weak network conditions. This was

resolved by adopting the MQTT protocol with the HiveMQ broker for lightweight, reliable publish-subscribe telemetry streaming.

2. WiFi disconnects. : Intermittent network drops caused data loss during continuous sensor streaming. This was resolved by implementing an automatic reconnection routine in the firmware to restore both WiFi and broker links seamlessly whenever a connection was lost.

6.5 Summary of findings

1. A low-cost wearable (ESP32 and MAX30102) was built, and it sends data through a cloud broker to a backend, a database, and a dashboard in real time.
2. A personalised BP model was trained on about 8 minutes of data from one person. It is clearly better than the baseline when tested on conditions similar to the training data (systolic MAE 4.91 mmHg, diastolic MAE 3.46 mmHg).
3. The performance drops on an activity that the model has not seen (systolic MAE 12.29 mmHg, R^2 0.026). The model is therefore not ready for general use.
4. The model mainly uses heart rate (79% importance), and the raw PPG signal adds little.
5. Message delivery was about 95% in a 1-minute test.
6. Fall risk is rule-based and was not tested on real falls or faints.
7. The system was tested only by the author, so the results are a proof of concept.

7 DISCUSSION

7.1 Interpretation of findings

BP prediction. The Random Forest model was much better than the baseline when it was tested on conditions similar to the training data. In the time-block test, the systolic MAE was 4.91 mmHg (baseline 11.65 mmHg), the diastolic MAE was 3.46 mmHg (baseline 7.25 mmHg), and both targets were inside the AAMI limits. This shows that, for one person, the model can learn the link between the sensor values and the cuff BP.

However, this result should be read with care. The 10-second blocks come from the same short recording of about 8 minutes, so neighbouring blocks have similar BP values. The dataset also has only 58 unique systolic and 41 unique diastolic values. The time-block result is therefore probably too optimistic.

Unseen activity. When the model was tested on an activity it had not seen, the systolic MAE rose to 12.29 mmHg and R^2 fell to 0.026. The systolic error was outside the AAMI limits (mean error 6.13 mmHg, SD 12.65 mmHg). The diastolic error stayed inside the limits (mean error 3.79 mmHg, SD 7.56 mmHg), but the SD was very close to the limit. This is the more honest test of how the model may behave in real use, and it shows that the model does not yet work well in new conditions.

What the model uses. The feature importance shows that the model mostly depends on the average heart rate (BeatAvg, 79%). The raw Red and IR values add only about 14%. This means that, in this dataset, the model mainly captures the link between heart rate and BP in this one person (for example, a higher heart rate during walking with a higher BP), and not the shape of the PPG waveform. This is a limit of the idea of “BP from PPG” in this study and not a proof that the PPG signal contains no BP information. Features taken from the waveform shape would be needed to test this

Fall risk. The fall risk status is calculated with threshold rules on the 5-second averages and on the 1-minute average (Normal, Risky, Critical). These rules were not tested on real falls or faints, because no such data was collected. The result shows that the system can raise a status from live data, but not that this status predicts falls.

System behaviour. The pipeline from the ESP32 to the broker, the backend, the database, and the dashboard worked in the tests. The delivery rate was about 87% in a 1-minute test 460 messages lost . For a monitoring system that is checked every few seconds, a small loss may be acceptable, but it must be measured over a longer time before any real use. The design also adds delays on purpose: the backend needs about 1.6 seconds of data before it gives the first heart rate, the dashboard values use a moving average of 10 values, the database record is a 5-second average, and the status uses a 1-minute average. These steps reduce noise and false alerts, but they also slow down the warning.

7.2 Comparison with literature baselines and existing solutions

Study	Data and method	Reported BP error	Compared with this work

7.3 Research contribution

- A personalised dataset of PPG, heart rate, and cuff BP values, collected with a low-cost sensor (MAX30102), and a clear description of how it was collected.
- A group-based evaluation method that avoids the wrongly high accuracy of a random split, and a comparison with a baseline and with the AAMI limits.
- An honest result: a personalised model can work for one person under similar conditions, but it does not generalise to an unseen activity, and the model relies mostly on heart rate.

- A design that links PPG-based vital signs to fall risk as an early-warning layer for a multi-component fall detection system.

7.4 Product contribution

- A working low-cost prototype: ESP32, MAX30102, battery, and a wristband box, with a battery level and charging status sent to the dashboard.
- A real-time pipeline: ESP32 → HiveMQ (MQTT over TLS) → FastAPI backend → Supabase → React dashboard, where more devices can be added by connecting them to the broker.
- A backend that filters the signal, detects the peaks, calculates heart rate and pulse interval, predicts BP, saves 5-second averages, and calculates a 1-minute status.
- A dashboard with live values and history views (recent, daily, monthly), and a patient event log that allows activities to be linked with the vital signs.
- A shared data interface that the alert component (Member 4) can use.

7.5 Limitations and threats to validity

Data and participants

- *One participant and about 8 minutes of data.* The model is personalised and may not work for other people. The results do not show how it behaves for elderly users.
- *Small number of independent BP values.* The cuff readings were taken about once per minute and added by hand, and each reading was applied to many PPG samples. This assumes that BP was stable during each measurement. The real number of independent samples is much smaller than 23,930.
- *Skin tone, tissue thickness, age, and health conditions* can change PPG signals, but they were not tested.

- *The reference.* The cuff itself has measurement error, and the PPG recording and the cuff reading were matched by timing only.

Model and features

- *Time-block validation is optimistic,* and the unseen-activity test shows weak results for systolic BP.
- *The model depends mostly on heart rate,* so it may not detect a BP change that is not linked to a heart rate change, such as a sudden BP drop when standing up.
- *Difference between training and deployment.* The training data used BPM and BeatAvg values from the ESP32 logging mode, while the deployed backend calculates them with a band-pass filter and peak detection. The model receives the instant heart rate and the average heart rate from the backend. If these values differ from the training values, the predictions may be affected.
- *PulseInterval_ms* in the dataset was a time-since-last-beat counter, so it was not used in the model. The pulse interval shown on the dashboard is calculated in the backend.

Fall risk

- *Rules were not tested on real falls or faints.* The thresholds come from references, not from this study's data. The link between BP or heart rate and falls was not tested here.
- *The false alarm rate was not measured,* so the benefit of the system for reducing false alarms is not proven.

System and security

- *Message loss.* About 10% of the messages were lost in a 1-minute test.
- *Network and laptop dependence.* The system needs internet, and the backend runs

- .*Security limits of the prototype.* The backend API allows requests from any origin, there is no login on the dashboard or the API, and all clients use broker credentials that must be kept secret.
- *Movement noise.* The finger sensor needs to be fixed firmly, and finger movement can add noise to the signal. This may be a problem for people with hand tremor or for long wearing.

Validation

- *No user or stakeholder validation.* The system was tested only by the author, not with elderly people or care staff.
- *The combined system was not evaluated as a whole.*

7.6 Implications

For the project. The physiological layer is technically connected to the rest of the system, so the alert component can use heart rate, BP, and status data. Before the data is trusted for real alerts, the model should be tested on more activities, and the false alarm rate should be measured together with the motion component.

For research.

- Personalised models are a possible way to handle the differences between people, but they need a calibration step with a cuff for each new user.
- Evaluation of PPG-based BP models should use group-based or unseen-condition tests. A random split of consecutive samples would give a wrongly high accuracy.

- Models should be checked to see whether they use the PPG waveform or only the heart rate.

For practice. The system is a research prototype. It must not be used to diagnose or treat, and it must not replace cuff measurements or the decisions of care staff. If it is developed further, care staff should treat the status as a hint that needs to be checked, and not as a decision. Some alerts could be false, and some risks could be missed.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This research aimed to build a low-cost, easy-to-wear PPG-based wearable component that estimates heart rate and blood pressure (BP) with a personalised model and gives a fall risk status, to support a complete elder-care fall detection system.

A prototype was built with an ESP32 and a MAX30102 sensor. The ESP32 sends raw Red and IR signals to a HiveMQ cloud broker over MQTT with TLS. A Python backend filters the signal, detects the heartbeats, calculates the heart rate and pulse interval, predicts BP with a Random Forest model, saves 5-second averages in a database, and sends live data to a React dashboard through WebSockets. The same data is available to the alert component of the system.

The BP model was trained on a personalised dataset of 23,930 samples from one participant, with a standard BP cuff as the reference. When it was tested on conditions similar to the training data, it was clearly better than a baseline that always predicts the mean (systolic MAE 4.91 mmHg compared with 11.65 mmHg, and diastolic MAE 3.46 mmHg compared with 7.25 mmHg), and both targets were inside the AAMI limits. However, when it was tested on an activity it had not seen, the systolic error rose to 12.29 mmHg and was outside the AAMI limits. The model also depends mostly on heart rate (79% importance), and the raw PPG values added little.

The fall risk status is calculated with threshold rules, because no falls or faints were recorded. It was not tested on real falls.

In conclusion, this work shows that a low-cost, real-time pipeline for personalised vital sign monitoring can be built and connected to a fall detection system. It is a proof of concept. The results do not show that the system predicts falls, that it works for other people, or that it is ready for use with elderly people.

8.2 Achievement of objectives

Objective	Status	Evidence
Collect raw PPG signals with the ESP32 and MAX30102 and calculate heart rate and pulse intervals	Achieved	The ESP32 sends raw Red and IR values. The backend calculates heart rate and pulse interval with filtering and peak detection (Sections 5.2, 6.2).
Build a personalised dataset using a standard BP cuff as the reference	Achieved, with limits	23,930 samples from one participant, with cuff BP added by hand (Section 3.2). The dataset is small.
Train and evaluate an ML model that predicts systolic and diastolic BP	Partly achieved	Good results on similar conditions, weak results on an unseen activity (Table 6.1, Table 6.2).
Classify fall risk using heart rate and BP	Partly achieved	Rule-based status (Normal, Risky, Critical) is implemented. It was not tested on real falls (Section 6.1.4).
Send data in real time through MQTT (HiveMQ) to the backend and dashboard	Achieved	The pipeline works. Delivery rate was about 87% in a 1-minute test , delay (Section 6.2).
Test the system and integrate it with the other components	Partly achieved	Integrate 3 systems as 1 using Hivemq cloud.
Build the BP model on a personalised dataset so that individual differences cause fewer false alerts	Not shown	A personalised model was built, but the effect on false alerts was not measured, and no comparison with a general model was made.
Evaluate how the fall risk information supports the overall fall detection decision	Not achieved	The combined system was not evaluated together with the motion and environment components. This is left for future work.

Research questions

- *RQ1 (How accurately can BP be predicted from MAX30102 PPG features?)* For one person, under similar conditions, the model reached a systolic MAE of 4.91 mmHg and a diastolic MAE of 3.46 mmHg. For an unseen activity, the errors were 12.29 mmHg and 7.34 mmHg. The model relied mostly on heart rate.
- *RQ2 (Can heart rate and predicted BP be used to classify fall risk reliably?)* The rules produce a status from live data, but reliability was not shown, because there is no fall or faint data.
- *RQ3 (Can the system send and show data in real time with low delay?)* Delivery rate was about 87% in a 1-minute test , delay (Section 6.2).
- *RQ4 (Does fall risk information improve fall detection reliability compared with motion data alone?)* This was not tested.

8.3 Individual research and product contributions

Product contributions

- A low-cost wearable prototype (ESP32, MAX30102, battery, wristband box) with battery level and charging status.
- A real-time pipeline with an MQTT cloud broker that allows more devices to be added without changing the backend.
- A FastAPI backend that filters the signal, calculates heart rate and pulse interval, predicts BP, saves records, and calculates a Normal, Risky, or Critical status.
- A React dashboard with live values and history views, and a patient event log.

- A shared data interface for the alert component.

8.4 Future directions

Data and model (most important)

1. **More data:** collect data over many days, at different times of the day, and during different activities, with many cuff readings, so that the number of independent BP values is much larger.
2. **More participants:** include people of different ages, skin tones, and health conditions, and finally elderly people, with ethics approval and consent. Test whether one model or a short personal calibration works best.
3. **Better features:** use waveform features of the PPG signal (for example, pulse shape and timing features) and not only heart rate, and check whether the model uses them.
4. **Fix the training and deployment difference:** train the model with the same heart rate values that the backend calculates, or move the same processing to the training script, so that the model sees the same kind of input in both cases.
5. **Better testing:** use day-wise tests (train on one day, test on another) and tests with new participants. Compare personalised and general models to see if personalisation reduces false alerts.
6. **Other models:** compare Random Forest with other methods when more data is available.

Fall risk

7. **Clinical thresholds:** confirm the thresholds with a clinician and with the literature.
8. **Real labels:** collect data about faints, dizziness, and low-BP events (for example, standing up quickly, orthostatic changes) with a clinician's help, and replace or improve the rules with a trained model.
9. **Combined evaluation:** test the full system with the motion and environment

components and measure whether the heart rate and BP information reduces false alarms and gives an earlier warning.

Hardware and system

10. **Wrist or other wearable forms:** a wrist or armband version would be more comfortable than a fingertip sensor, but it needs better motion noise handling. An accelerometer could be used to detect and remove movement noise.
11. **Battery and size:** measure and improve battery life, and reduce the size.
12. **Reliability:** improve message delivery (for example, a higher MQTT quality-of-service level and local buffering when WiFi is lost) and measure the delay and the loss over long tests.
13. **Cloud deployment:** run the backend on a server and not on a personal laptop, and support many patients and devices with separate topics and accounts.
14. **Security and privacy:** add login and roles for the dashboard and the API, restrict the allowed origins, use a separate credential for each device, store secrets outside the code, and encrypt and protect the stored health data.

Product and ethics

15. **User studies:** test comfort, usability, and the usefulness of alerts with elderly people and care staff.
16. **Regulation:** follow the rules for medical devices and health data before any real use, and validate with the full AAMI/ISO protocol if BP accuracy is to be claimed.